ALLEN® CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

ENTHUSIAST ADVANCE COURSE

PHASE : MEA-I (A)

TARGET : PRE-MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 05-02-2023

ANSWER KEY

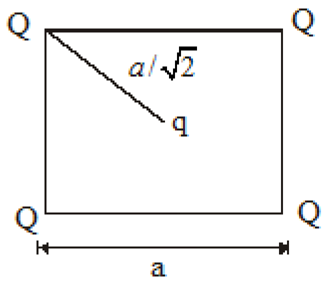
Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	2	4	1	2	3	2	2	2	2	3	3	2	3	2	4	2	2	3	4	2	1	3	4	3	1	3	2	1	2
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	1	2	1	1	2	3	1	2	2	4	3	4	2	3	2	1	1	2	2	1	2	2	4	2	2	3	3	2	3	2
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	4	2	1	3	4	2	3	1	2	1	1	1	4	3	2	4	1	3	4	2	2	4	1	4	4	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	2	2	3	3	1	1	3	2	2	2	2	2	2	3	3	4	4	1	1	2	4	1	3	2	3	2	1	4	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	3	4	4	4	1	4	4	4	3	3	2	4	3	2	4	3	2	2	2	1	2	3	1	2	1	3	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	3	2	4	1	2	3	1	4	4	1	1	2	1	1	3	2	2	1	4	3	3	1	2	3	4	3	4	3	2	2
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	1	1	2	2	2	4	1	2	1	1	1	2	2	2	4	4	1	3	2	3										

HINT – SHEET

SUBJECT : PHYSICS

SECTION - A

12. Ans (3)

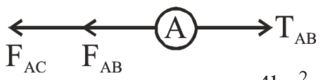


F_{Net} on Q will be zero

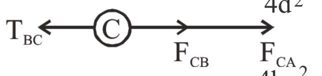
$$\text{if } \frac{k\theta^2}{a^2}\left(\sqrt{2}+\frac{1}{2}\right)=\frac{-kQq}{\left(\frac{a}{\sqrt{2}}\right)^2}$$

$$q=-\frac{Q}{2}\left(\sqrt{2}+\frac{1}{2}\right)$$

15. Ans (2)



$$T_{AB}=F_{AC}+F_{AB}=\frac{4kq^2}{4d^2}+\frac{2kq^2}{d^2}=\frac{3kq^2}{d^2}$$



$$T_{BC}=F_{CA}+F_{CB}=\frac{4kq^2}{4d^2}+\frac{8kq^2}{d^2}=\frac{9kq^2}{d^2}$$

$$\boxed{\frac{T_{AB}}{T_{BC}}=\frac{1}{3}}$$

16. Ans (4)

$$\vec{F}=\frac{K.q_1q_2\vec{r}}{|r|^3}$$

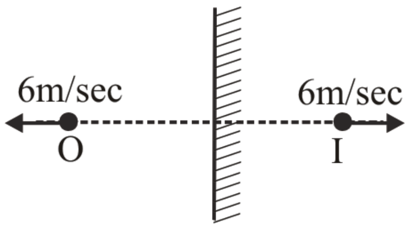
$$=-\frac{1}{4\pi\epsilon_0}\times\frac{q_1.q_2(2\hat{i}-\hat{j}+3\hat{k})}{\left[\sqrt{2^2+(-1)^2+(3)^2}\right]}$$

17. Ans (2)

Dielectric constant $k=\frac{\varepsilon}{\varepsilon_0}$ Permittivity of metals (ε) is assumed to be very high.

19. Ans (3)

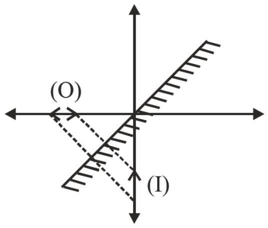
Relative speed of image w.r.t. object = 6 – (–6) = 12 m/sec.



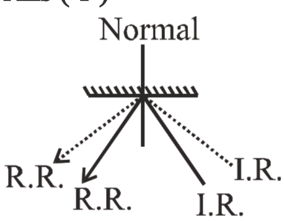
20. Ans (4)

$\frac{5}{2} = 2.5$ feets

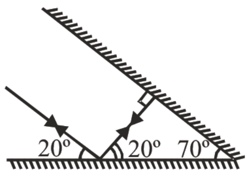
21. Ans (2)



22. Ans (1)



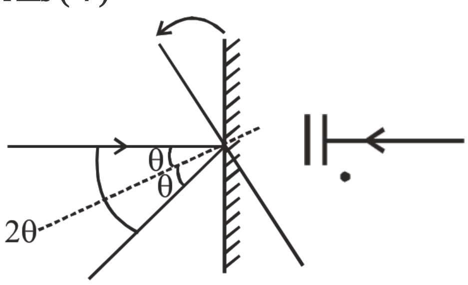
23. Ans (3)



$\delta = 180°$

Ray retraces its path

24. Ans (4)



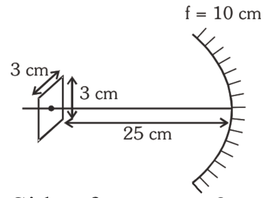
25. Ans (3)

Here focal length = f and u = –f

On putting these values in

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow \frac{1}{f} = -\frac{1}{f} + \frac{1}{v} \Rightarrow v = \frac{f}{2}$$

26. Ans (1)



Side of square = 3cm

$A_0 = 3 \times 3 = 9 \text{ cm}^2$

$A_I = m^2 \times A_0$

$$m = \frac{f}{f-u} = \frac{-10}{-10-(-25)} = \frac{-10}{15} = \frac{-2}{3}$$

$$A_I = \left(\frac{-2}{3}\right)^2 \times 9 = 4 \text{ cm}^2$$

27. Ans (3)

$$\frac{AD}{40} = \frac{1}{4/3}$$

AD = 30 cm

28. Ans (2)

Distance of object is equal to focal length

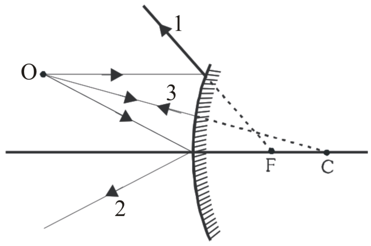
So, $u = \pm f$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{\pm f} = \frac{1}{f}$$

$v = 2f, \infty$

29. Ans (1)

Correct Ray diagram is :



30. Ans (2)

$(v_I)_z = 2(v_M)_z - (v_O)_z$

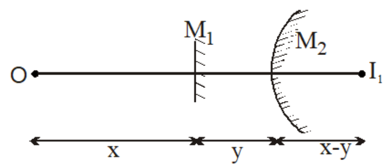
$(v_I)_z = 2(0) - (3\hat{k})$

$(v_I)_z = -3\hat{k}$

$v_I = (v_I)_x + (v_I)_y + (v_I)_z$

$v_I = \hat{i} + 2\hat{j} - 3\hat{k}$

31. Ans (1)



$$\frac{1}{x-y} + \frac{1}{-(x+y)} = \frac{1}{f} \Rightarrow f = \frac{x^2 - y^2}{2y}$$

Now, as this is convex mirror, f is (+)ve hence magnitude of $f = \frac{x^2 - y^2}{2y}$

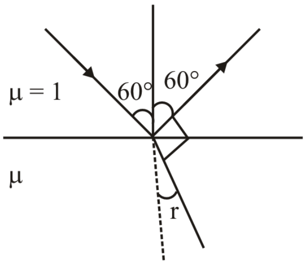
32. Ans (2)

Distance $= h_1 + \frac{h_2}{\mu}$

33. Ans (1)

$\lambda_{medium} = \frac{\lambda_{air}}{\mu} = \frac{6000}{1.5} = 4000\overset{o}{A}$

34. Ans (1)



$60 + r = 90°$

$r = 30°$

$1\sin 60° = \mu \sin 30°$

$\boxed{\mu = \sqrt{3}}$

35. Ans (2)

From Snell's law $\quad r = 2C'$

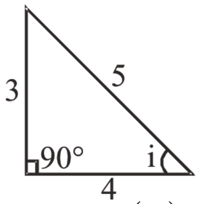
$1.6 \times \sin i = 1 \times \sin r$

$1.6 \times \sin i = \sin 2C'$

$1.6 \times \sin i = 2 \sin i \times \cos i$

$\cos i = \frac{1.6}{2} = \frac{4}{5}$

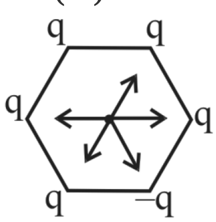
$\sin i = \frac{3}{5}$



$i = \sin^{-1}\left(\frac{3}{5}\right)$

SECTION - B

38. Ans (2)



$F_{net} \neq 0$

40. Ans (4)

Due to the symmetry of charge. Net Electric field at centre is zero.

42. Ans (4)

Let r be the distance between q_1 and q_2.

According to Coulomb's law, the force between them is,

$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$

$= \frac{1}{4\pi\varepsilon_0}\frac{q_1(q - q_1)}{r^2} \qquad (\because q_1 + q_2 = q)$

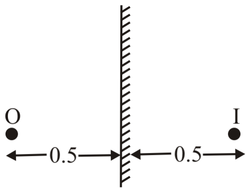
For F to be maximum $\frac{dF}{dq_1} = 0$ as q and r are constants.

$\therefore \frac{d}{dq_1}\left[\frac{1}{4\pi\varepsilon_0 r^2}(q_1 q - 1_1^2)\right] = 0$

$q - 2q_1 = 0 \qquad \left[\text{As } \frac{1}{4\pi\varepsilon_0 r^2} \neq 0\right]$

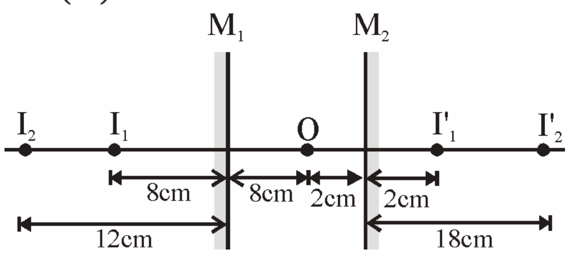
$\frac{q_1}{q} = \frac{1}{2} = 0.5$

43. Ans (2)



distance between object and image = 0.5 + 0.5 = 1 m

44. Ans (3)



$M_2I'_1 = 2$ cm, $M_2I_1 = M_2I'_2 = 18$ cm

$M_2I_2 = 10 + 12 = 22$

45. Ans (2)

$n_{ga} = \frac{3}{2}, n_{wa} = \frac{4}{3}$

$\therefore n_{gw} = \frac{n_{ga}}{n_{wa}} = \frac{3/2}{4/3} = \frac{9}{8}$

46. Ans (1)

$R = +20$ cm

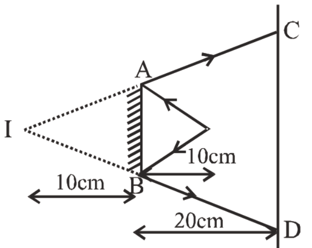
$\Rightarrow f = \frac{R}{2} + 10\text{cm}; u = -14\text{cm}$

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$

$= \frac{1}{10} + \frac{1}{14} = \frac{7+5}{70} = \frac{12}{70}$

$\therefore v = \frac{70}{12}$cm

47. **Ans (1)**



Δ^S ABI & CDI are similar

$\frac{AB}{CD} = \frac{10}{30} = \frac{1}{3}$

CD = 3(AB) = 3(20)

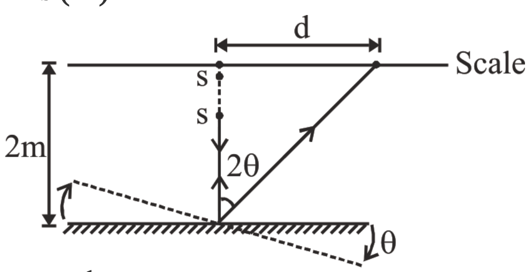
CD = 60 cm

time $= \frac{\text{Dis tan ce}}{\text{Speed}} = \frac{CD}{10} = \frac{60}{10} = 6$ sec.

48. **Ans (2)**

Apparent height $= \mu h = \frac{4}{3} \times 24 = 32$ cm

49. **Ans (2)**



$2\theta = \frac{d}{2}$

$2 \times 1.8° = \frac{d}{2}$

$d = 4 \times 1.8°$

$d = 4 \times 1.8 \times \frac{\pi}{180} = 0.1256$ m

50. **Ans (1)**

For concave mirror

u = +120 cm

f = –60 cm

$v = \frac{uf}{u-f} = \frac{(120)(-60)}{120-(-60)} = -40$ cm

$m_1 = -\frac{v}{u} = -\frac{-40}{120} = +\frac{1}{3}$

For convex mirror :

u = +(40 – 15) = +25 cm

f = +30 cm

$m_2 = \frac{f}{f-u} = \frac{30}{30-25} = +6$

$m_{net} = m_1 \times m_2 = \frac{1}{3} \times 6 = +2$

SUBJECT : CHEMISTRY

SECTION-A

51. **Ans (2)**

The order of shielding effect of various orbital electrons is s > p > d > f. Due to the poor shielding effect of 4f- electrons in 5d-series elements, there is enhanced increase in effective nuclear charge. As a result of this the valence electrons are tightly bound with the nucleus and thus their removal require higher energy.

59. **Ans (3)**

electronegativity (approx. values)

F 4.0

Cl 3.16

C 2.5

P 2.19

H 2.2

H - P = 2.2 - 2.19 = 0.01

60. **Ans (2)**

NCERT Pg. # 85

61. **Ans (3)**

NCERT Pg # 90

62. **Ans (3)**

NCERT Pg # 76

63. **Ans (2)**

NCERT Pg # 82

71. **Ans (1)**

From the given data half life is directly proportional to initial concentration or pressure therefore order of reaction is zero.

76. **Ans (4)**

$ROR = \frac{-d[A]}{\frac{dt}{2}} = \frac{-[0.3-0.8]}{5 \times 2}$

$\frac{0.5}{5 \times 2} = \frac{0.1}{2} = 5 \times 10^{-2}$ M.min^{-1}

NCERT Pg. # 98

77. **Ans (3)**

NCERT Pg. # 98

78. **Ans (2)**

NCERT Pg. # 106

79. **Ans (4)**
NCERT Pg. # 103, 104, 112

81. **Ans (3)**

$$\log\left(\frac{K_2}{K_1}\right) = \frac{Ea}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

$$\log\left(\frac{4K_1}{K_1}\right) = \frac{Ea}{2.303 \times R}\left[\frac{1}{290} - \frac{1}{310}\right]$$

$$\frac{0.60 \times 2.303 \times R \times 310 \times 290}{20} = Ea$$

$$\frac{1.38 \times R \times 310 \times 290}{20} = Ea$$

82. **Ans (4)**

$$t_{99.9\%} = 10 \times t_{1/2}$$

$$t_{1/2} = \frac{60}{10} = 6 \text{ min}$$

84. **Ans (2)**

$$K = \frac{2.303}{t}\log\frac{100}{100 - x}$$

$$\frac{0.693}{693} = \frac{2.303}{t}\log\left(\frac{100}{100 - 90}\right)$$

$$10^{-3} = \frac{2.303}{t}\log\left(\frac{100}{10}\right)$$

$$t = \frac{2.303}{10^{-3}} = 2303 \text{ years.}$$

85. **Ans (4)**

$$ROR\ (r) = K = \frac{(ROA)\ H_2}{3}$$

$$(ROR)_{H_2} = 3 \times 5 \times 10^{-4}$$

$$= 15 \times 10^{-4} \text{ mol } L^{-1}\ S^{-1}$$

SECTION-B

86. **Ans (1)**

$$X_{(g)} \xrightarrow[-e^-]{IP_1} X^+ \xrightarrow[-e^-]{IP_2} X^{2+}$$

90. **Ans (2)**
NCERT Pg # 89

93. **Ans (2)**

$$= \frac{t_{75\%}}{t_{50\%}} = \frac{\frac{2.303}{k}\ \frac{100}{25}}{\frac{2.303}{k}\ \frac{100}{50}}$$

$$t_{75\%} = 2 \times 16 = 32$$

94. **Ans (3)**
Threshold energy = activation energy + energy possessed by reactant
= 32 + 18 = 50 K cal/mol

95. **Ans (3)**
Ref. NCERT

99. **Ans (2)**

$$\frac{r_2}{r_1} = (2)^{\frac{40-10}{10}}$$

$$\frac{r_2}{r_1} = 2^3 = 8$$

100. **Ans (2)**
NCERT Pg. # 107

SUBJECT : BIOLOGY-II

SECTION-A

155. **Ans (2)**
NCERT XII, Pg # 23

156. **Ans (3)**
NCERT XII Pg.# 24

159. **Ans (4)**
NCERT-XII, Pg#21

160. **Ans (1)**
NCERT(XII) Pg # 21 (E), 22 (H)

171. **Ans (3)**
NCERT [XII] pg. 29

172. **Ans (1)**
NCERT (XI) Pg. # 85

175. **Ans (4)**
NCERT XII pg.# 34

178. **Ans (3)**
NCERT XII page 34-35, C & D are true

SECTION-B

187. **Ans (1)**
NCERT-XII, Pg#23, Para.-II

188. **Ans (2)**
NCERT XII, Pg # 23

194. **Ans (2)**
NCERT (XII) Pg # 34

195. **Ans (4)**
NCERT–XII, Pg. # 34

197. **Ans (1)**
NCERT XII Pg.# 35

200. **Ans (3)**
Microspore, Generative cell, synergid, antipodal cell, vegetative cell of pollen